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RESEARCH ARTICLE

A Synergistic Stacked Ensemble Deep Learning Model for Predicting Diabetic Retinopathy Severity

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ABSTRACT Diabetic retinopathy is a leading cause of vision loss worldwide. Early and accurate diagnosis can prevent vision loss, and automatic grading is particularly important to assist physicians in making timely decisions. This work proposes an efficient stacked ensemble framework for the DR severity classification task. Unlike traditional ensembles, our model leverages out-of-fold predictions and collaborative meta-learning to improve ordinal consistency without leaking information during training. We incorporate the best features of three top deep learning architectures: Swin-B Transformer, EfficientNetV2-L, and ConvNeXt-L. A meta-learner maximizes their collective capacity for prediction. Extensive experiments on two widely available datasets, the APTOS 2019 Blindness Detection dataset and a larger Kaggle dataset, were conducted to evaluate our approach. We investigated the generalization capability of our ensemble with both 3-fold and 5-fold cross-validation. With the highest classification accuracy of 97.9% for the APTOS dataset and 98.2% for the Kaggle dataset, the results clearly indicate that our stacked model outperforms all individual networks. Furthermore, for both datasets, we consistently achieved a high QWK value of 0.97. This reflects strong agreement between our predictions and professional clinical evaluations. It also demonstrates the goodness of our model dealing with ordinal data. These findings emphasize that there is great potential in Stacked Ensembling to enhance the accuracy and robustness of automated diabetic retinopathy grading systems, particularly when combined with diverse high performance base models.

INDEX TERMS Diabetic retinopathy grading, stacked ensemble, deep learning, convolutional neural networks, transformer networks.

I. INTRODUCTION

Diabetic Retinopathy (DR) is a common chronic eye disease that leads to blindness. The DR is involved in five stages as per the Early Treatment Diabetic Retinopathy Study (ETDRS), which is a significant contributor to preventing blindness [1]. The first stage of the DR is Stage 0, which means the fundus image is normal and has no DR. The second stage is Stage 1, which means mild DR is characterized by the microaneurysms. The third stage is Stage 2, which means moderate is characterized by hemorrhages, hard exudates, blood vessels, and venous beading. Fourth stage is Stage 3, which means Severe is characterized by the intraretinal microvascular abnormalities and extensive hemorrhages.

The last stage is Stage 4, which means Proliferative DR is characterized by the retinal neovascularization and vitreous hemorrhage has shown in Figure 1. The clinical need for accurate staging is complicated by class imbalance in datasets and the varying feature sets required for different stages, ranging from identifying microaneurysms in early disease to spatial reasoning across the retina in advanced cases. Recent advances in deep learning have made

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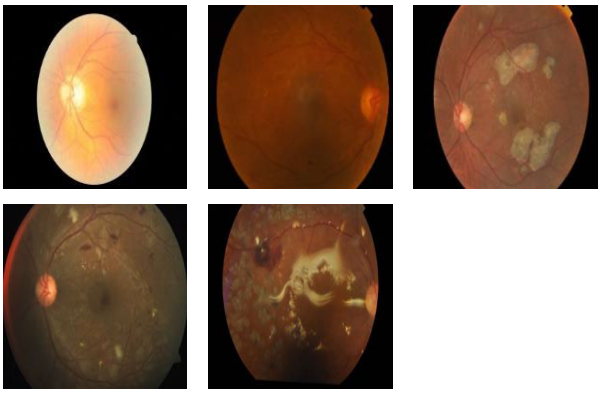


FIGURE 1. (a) Normal, (b) Mild, (c) Moderate, (d) Severe, (e) Proliferative DR.

significant strides in solving these diagnostic problems. Since 2020, changes to architecture have greatly improved the accuracy of automated DR detection. Through hierarchical self-attention mechanisms that capture long-range dependencies in fundus images, transformer-based models—like the Swin Transformer—have set new standards and are especially skilled at recognizing global pathological patterns typical of advanced stages [2]. The authors detected the early diseases using convolutional neural networks such as ConvNeXt and EfficientNetV2, and extracted features from the models that can detect the microvascular anomalies from the fundus images [3], [4].

The problems occur for a variety of patient populations with generalizability, interpretability, and computational efficiency. Prospective solutions are provided by recent developments in cross-modal learning and uncertainty-aware ensemble designs, especially when it comes to resolving dataset shift, a recurring problem in medical imaging [5]. Combining these cutting-edge computational methods with clinical staging systems is a big step toward automated, precise, and easily accessible DR screening solutions. It also opens the door to more scalable and trustworthy ophthalmology diagnostic tools. Among working-age adults worldwide, diabetic retinopathy (DR) continues to be a major preventable cause of blindness. Around 103.12 million adults globally suffered from DR as of 2020, and estimates suggest that number will rise to 160.50 million by 2045. Adherence to DR screening programs is suboptimal, especially in low- and middle-income countries where the screening adherence rate is roughly 39.3%, as opposed to 66.9% in high-income countries, despite the critical importance of early detection. In contrast to previous ensemble studies, this work improves ordinal reliability in DR grading by introducing a stacked meta-learning strategy that combines transformer-based and convolutional architectures with fold-wise non-leakage validation.

The following are the main goals of the work:

- Design an enhanced meta-learning fusion approach that makes use of out-of-fold (OOF) predictions from complementary architectures (Swin-B Transformer, EfficientNetV2-L, and ConvNeXt-L).
- Evaluate the performance evaluation using 3-fold and 5-fold cross-validation to show strong generalization across two different and heterogeneous datasets (APTOS 2019 and Kaggle).
- Develop the model with high consistency using ordinal consistency with Quadratic Weighted Kappa (QWK) metrics to ensure compliance with clinical grading guidelines.

The remainder of the manuscript is divided into the following sections: Existing relevant work is presented in Section II, and the Stacking Model for DR is proposed in Section III. Section IV presents the performance evaluation of the proposed method. Finally, Section V provides the conclusion.

II. LITERATURE SURVEY

A diabetic retinopathy detection method was proposed using image resizing, feature extraction with Haralick texture features and Hu moments, and classification via machine learning algorithms, including SVM, Decision Tree, K-NN, Naive Bayes, and EfficientNet. The EfficientNet model achieved the highest accuracy of 91.8% on a Kaggle dataset [6]. A method for identifying diabetic retinopathy from fundus images, using a neural network to diagnose micro-aneurysms, was proposed and showed that the proposed Support Vector Machine (SVM) classifier outperformed a Naive Bayes classifier with improved sensitivity (SVM: 58%, Naive-Bayes Method: 54%), specificity (SVM: 42%, Naive-Bayes Method: 44%), and accuracy (SVM: 53%, Naive-Bayes Method: 50%) compared to the Naive Bayes method [7]. An automated diabetic retinopathy detection approach using a convolutional neural network (CNN) was presented, achieving average accuracies of around 73.83% for training and 71.96% for validation on a publicly available dataset [8]. A deep learning system utilizing the LeNet-5 convolutional neural network (CNN) architecture was developed for diabetic retinopathy detection and stage classification, employing preprocessing techniques such as edge detection, thresholding, and data augmentation. The authors used a retinal images dataset, and the model generated an average accuracy of 97%, a recall of 22% and an F1-score of 65% [9]. The authors proposed a hybrid model for DR using XGBoost for classification, Inception V3 for feature extraction from fundus images, and for feature selection, the Particle Swarm Optimization and Artificial Bee Colony were used. This model was implemented on the Messidor-2 dataset and got an accuracy of 93.1% [10]. To identify and severity of diabetic retinopathy from fundus images and data augmentation using a convolutional neural network (CNN) was used. This CNN model trained on the Kaggle dataset got an accuracy of 79.49% [11]. The authors proposed a hybrid model for detecting the grading of DR in a resource-constrained environment. In this paper, they used the EfficientNet model for feature selection and applied an ensemble boosting algorithm for classification. The model got an accuracy of 99.1% for Efficient and 99.2% for XGBoost combined on the APTOS

and IDRID datasets, respectively [12]. The ResNet-50 CNN model is used for detecting the DR from the fundus images, achieving an accuracy of 82%, precision of 81%, recall of 82% and F1-score of 81% on the Messidor dataset [13]. The InceptionV3 pre-trained model is used for detecting DR applied on EyePACS and APTOS 2019 datasets. The model generated an accuracy of 81.61% and an F1-score of 80.21% on the APTOS 2019 dataset where whereas in the EyePACS dataset got an accuracy of 74.28% and an F1-score of 73.81% [14]. A transfer learning approach is used to classify the DR on the APTOS 2019 dataset using improved ResNet50. It detects the severity of DR and generates an accuracy of 90% and a Cohen's Kappa of 94% [15]. The authors proposed detecting DR using an ensemble learning approach on Messisor dataset. In this papers the authors used support vector machines, random forest, Naïve Bayes and decision trees, the model achieved an overall accuracy of 68.35%, the SVM classifier got an accuracy of 77.48% for individual models [16]. The proposed model hybrid Residual U-Net (HRUNET) that combines residual modules and skip connections for detecting the severity of DR. The model got an accuracy of 91% on KAGGLE dataset and 94% on APTOS dataset [17]. a hybrid approach (HCF+CNN) for diabetic retinopathy screening and referral using the MESSIDOR dataset. Their combined method yielded an AUC of 0.8932 with a sensitivity of 0.9109 and an AUC of 0.9347 with a sensitivity of 0.9721 for DR screening [18]. CNNs used to detect diabetic retinopathy have training accuracies that vary depending on the stage of the disease, from 90% for proliferative DR to 99.5% for no DR. But the validation accuracy ranged more significantly, from 85% for No DR to 55% for Proliferative DR, underscoring the difficulties in extrapolating across severity levels [19]. By incorporating Inception modules into a U-Net framework, a novel architecture known as Hybrid Inception U-Net (HIUNET) was put forth for the automated diagnosis of diabetic retinopathy. Using the APTOS and Kaggle datasets, the HIUNET model was assessed; the accuracy rates were 95% and 92%, respectively [20], using the APTOS2019 and Diabetic Retinopathy Competition datasets to train the AlexNet and DenseNet-169 architectures. For normal, mild, moderate, severe, and Proliferative Diabetic Retinopathy (PDR) stages, DenseNet-169 obtained F1-scores of 0.55, 0.38, 0.40, 0.60, and 0.69, respectively [21]. On the publicly available dataset, a hybrid deep learning method for predicting diabetic retinopathy that combined the DenseNet201 and Xception models with a stacking technique produced an overall accuracy of 94.67% [22]. A diabetic retinopathy classification system that combines ResNet-18 for feature extraction with the Slime Mould Algorithm (SMA) for feature selection to enhance classification performance. Using a Kaggle dataset, the system achieved a classification accuracy of 98.40%, demonstrating the effectiveness of the SMA-based optimization [23]. A multi-label deep learning model for diabetic retinopathy (DR) classification pinpoints lesion regions using Grad-CAM. The method achieved a sensitivity of 93.9% and a specificity of 94.4% in DR classification on the test images [24]. An intelligent machine learning-based diabetic retinopathy grading and classification (IML-DRGC) model, aiming for automated diagnosis using IoT-captured retinal images. The IML-DRGC model achieved a sensitivity of 94.27%, a specificity of 96.17%, and an accuracy of 95.71% [25]. The authors proposed detecting diabetic retinopathy using adversarial attacks and validated the model on a real-world public DR dataset [26]. In this paper the authors proposed a Ensembling machine learning techniques for automatically detection as well as the stages of DR and implemented on KAGGLE dataset with an accuracy of 98.35% [27]. The authors proposed an improved feature extractor that improves feature discrimination across adjacent grades by lowering prediction inconsistencies between asymmetric bi-classifiers [28]. The authors proposed the detection of diabetic retinopathy into two stages (binary classification) and four stages (multi-class classification) using a deep neural network called RSG-Net (Retinopathy Severity Grading) [29]. The authors proposed to improve the detection of DR using a novel framework which combines the hybrid convolutional neural network – redial basis function classifier with multi-scale discriminative robust local binary pattern features [30].

III. METHODOLOGY

A. ABOUT DATASET

This study utilizes two publicly available and widely benchmarked diabetic retinopathy (DR) datasets: the APTOS 2019 Blindness Detection dataset and the Kaggle EyePACS (2015) Diabetic Retinopathy Detection dataset. Both datasets are licensed under the Creative Commons Attribution-Non-Commercial-Share Alike 4.0 International License (CC BY-NC-SA 4.0) and were accessed from Kaggle ([31], [32]).

1) APTOS 2019 BLINDNESS DETECTION

This dataset, released by the Asia Pacific Tele-Ophthalmology Society (APTOS), contains 3,662 color fundus images acquired from 1,831 unique patients, with images captured for both left and right eyes.

2) KAGGLE EYEPACS (2015)

The dataset consists of 88,702 fundus images. In this work the authors had taken 35,126 images. The labels employ KAGGLE five DR severity grades.

The cross-validation folds (k=3 and k=5) were used to ensure reproducibility, independence and patient-level stratification. Two eyes of every patient were consistently allocated to the split. This process eliminates the eye data leakage and ensures fair model evaluation.

B. PROPOSED SYNERGETIC ENSEMBLE STACKING CLASSIFIER FOR THE PREDICTION OF DIABETIC RETINOPATHY

This study's methodology is based on a stacked ensemble learning framework, which combines the predictive

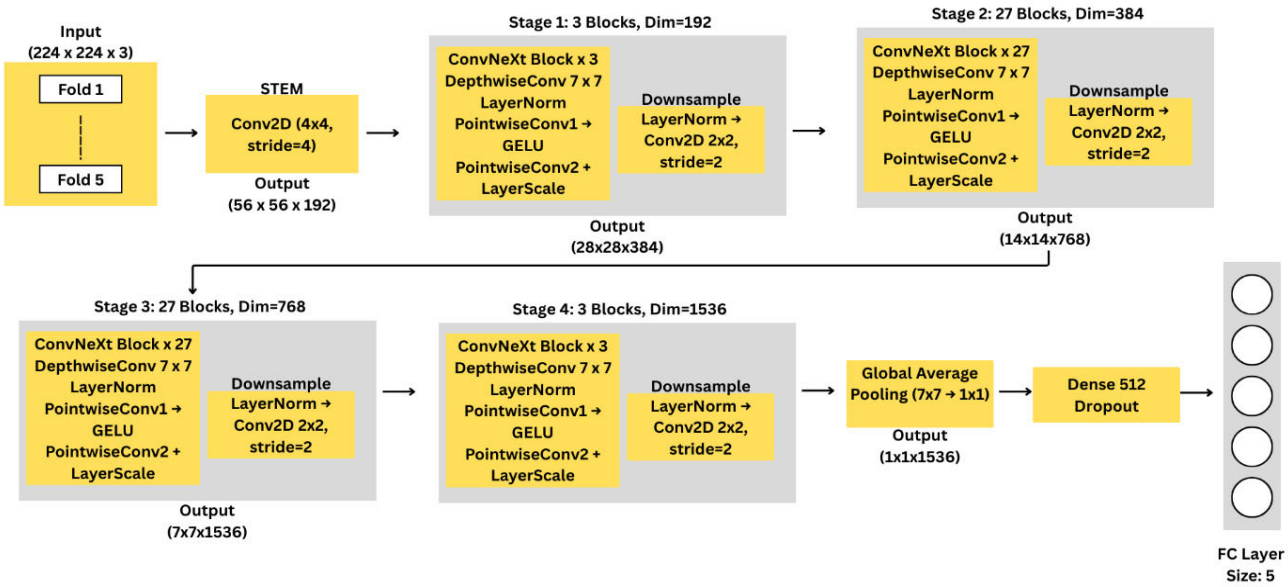


FIGURE 2. ConvNeXt-L architecture.

capabilities of several different base models to produce reliable and accurate classification. Every image was normalized, resized to 512 × 512 pixels, and center cropped. Random flips, rotations, and brightness adjustments were all used to augment the data. To address class imbalance, models were trained with weighted categorical cross-entropy using cosine annealing scheduling and the AdamW optimizer (LR $= 1 \times 10^{-4}$). Training was carried out using the Keras framework and TensorFlow-GPU on a 64GB computer equipped with an Intel® Xenon® Platinum 8276 CPU@2.20GHz and a 4GB NVIDIA GRID V100D-4Q graphics card. Random seeds were fixed for reproducibility. ConvNeXt-L, EfficientNetV2-L, and Swin-B Transformer are the three unique state-of-the-art deep learning architectures that make up the ensemble. A 5-fold cross-validation technique is initially used to train these base models independently. The two main functions of this cross-validation method are to produce out-of-fold (OOF) predictions for the complete training dataset and to give a more accurate estimate of each model's generalization performance. These OOF predictions serve as the input features for the next meta-learning step, in which a model that has not been trained on that data point makes the prediction for each data point. An essential base model for the stacking ensemble is the ConvNeXt-L architecture, which is depicted in Figure 2. Learning various hierarchical representations from the input images is made possible by its design, which consists of a stem layer followed by four stages of ConvNeXt blocks with progressive down sampling and increasing feature dimensions $X \in \mathbb{R}^{224\times224\times3}$ To improve the robustness of the learned features, the input is first divided into overlapping folds (Fold 1 to Fold 5). After that, the stem layer applies a 2D convolution, which is represented by equation (1).

$$Y_0 = Conv2D\left(X, W_0, b_0, stride = 4\right) \tag{1}$$

where $W_0 \in \mathbb{R}^{4\times4\times4\times192}$ and $b_0 \in \mathbb{R}^{192}$, yielding an output $Y_0 \in \mathbb{R}^{56\times56\times192}$.

The network then uses four stages. In these stages, every ConvNeXt block carries out a number of transformations. equation (2) can be used to depict a depthwise convolution.:

$$Z = DepthwiseConv2D\left(Y_{i-1}, W_{dw}\right) \tag{2}$$

with $W_{dw} \in \mathbb{R}^{7\times7\times C_i\times1}$. Layer Normalization follows in equation (3):

$$\hat{Z} = LN\left(Z\right) \tag{3}$$

A pointwise convolution then adjusts the feature dimension represented in equation (4):

$$A = Conv2D\left(\hat{Z}, W_{pw1}, b_{pw1}\right) \tag{4}$$

where $W_{pw1} \in \mathbb{R}^{1\times1\times C_i\times4C_i}$. The non-linear GELU activation can be represented in equation (5):

$$B = GELU\left(A\right) = A \cdot \Phi\left(A\right) \tag{5}$$

Another pointwise convolution follows in equation (6):

$$C = Conv2D(B, W_{pw2}, b_{pw2}) \tag{6}$$

With $W_{pw2} \in \mathbb{R}^{1\times1\times C_i\times4C_i}$. Finally, Layer Normalization and a residual connection are applied in equation (7).

$$Y_i = LN\left(C\right) + Y_{i-1} \tag{7}$$

Downsampling between the stages (except the last) is achieved using the equation (8):

$$Y_{down} = Conv2D(LN\left(Y_{in}\right), W_{down}, b_{down}, stride = 2) \tag{8}$$

where $W_{down} \in \mathbb{R}^{2\times2\times C_{in}\times C_{out}}$.

The specific configurations of ConvNeXt blocks in each stage are 3 (Stage 1, Dim=192), 27 (Stage 2, Dim=384),

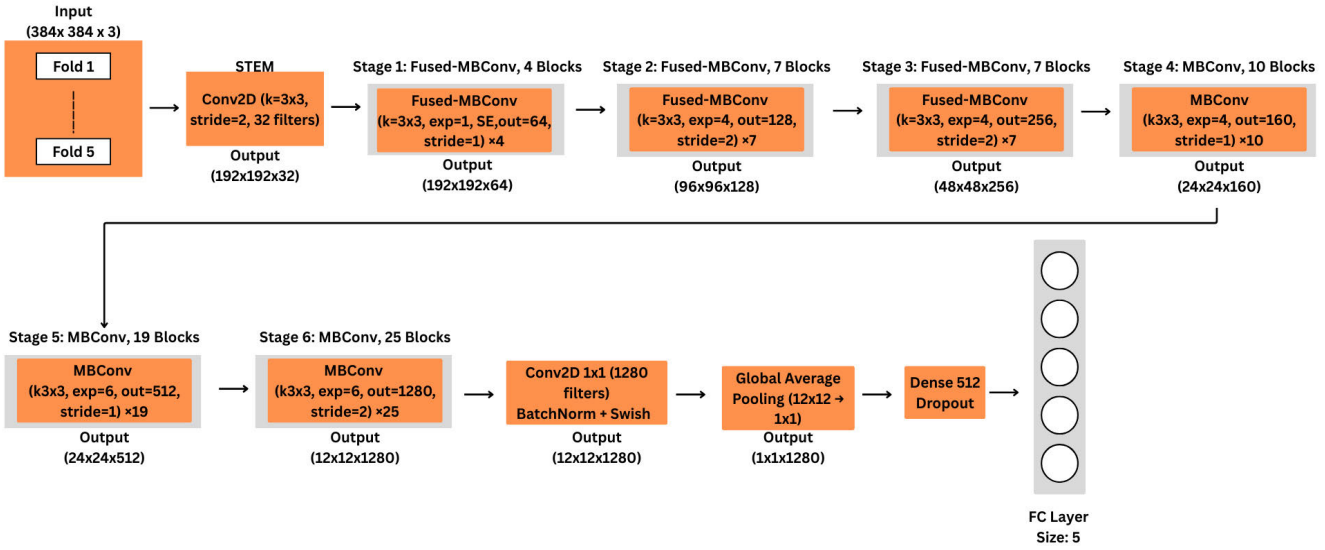


FIGURE 3. EfficientNetV2-L architecture.

27 (Stage 3, Dim=768), and 3 (Stage 4, Dim=1536), with corresponding output feature map sizes of $28\times24\times384$, $14\times14\times768$, *and* $7\times7\times1536$ After the downsampling layers.

Following Stage 4, a global average pooling layer aggregates the spatial information that can be represented in equation (9):

$$\vartheta = GAP\left(Y_{stage4}\right) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W} Y_{stage4}(i,j) \tag{9}$$

resulting in a feature vector $\vartheta = \mathbb{R}^{1536}$. This vector is then processed by a dense layer with 512 units and a dropout layer to avoid overfitting. Ultimately, a fully connected layer with Softmax activation generates the classification probabilities represented in equation (10):

$$p = Softmax(W_{fc}\cdot Dropout\left(Dense\left(\vartheta\right)\right) + b_{fc}) \tag{10}$$

where $p \in \mathbb{R}^5$ represents the output of this base model, these predicted probabilities will be utilized as input features for the meta-learner in the subsequent stacking process, leveraging the diverse and hierarchical feature representations learned by this ConvNeXt-based architecture. The EfficientNetV2-L architecture depicted in Figure 3 serves as a second distinct base model. Like the first base model, it begins by partitioning the input image into X elements of double cap R, resulting in one image. $X \in \mathbb{R}^{384\times384\times3}$ Into five overlapping folds. These folds are then processed by a stem layer consisting of a 2D convolutional layer, which can be represented in equation (11): where $W_0 \in \mathbb{R}^{3\times3\times3\times32}$ and $b_0 \in \mathbb{R}^{32}$, resulting in an output $Y_0 \in \mathbb{R}^{192\times192\times32}$.

$$\begin{aligned} Y_0 &= Conv2D(X.W_0, b_0, kernel \\ &= 3\times3, stride = 2, filters = 32) \end{aligned} \tag{11}$$

Stage 2 incorporates 7 Fused-MBConv blocks with a kernel size of $k = 3\times3$, an expansion factor of $exp = 4$, output channels of $out = 128$, and a stride of $s = 2$. This stage includes a Squeeze-and-Excitation (SE) module. The Fused-MBConv operation can be generally described in equation (12):

$$\begin{aligned} &Y_{i,fused} \\ &= Conv2D(Conv2D\left(Y_{i-1}, W_{exp}, b_{exp}\right), W_{out}, b_{out}, kernel \\ &= k, stride = s) \end{aligned} \tag{12}$$

where W_{exp} expands the channels and W_{out} projects them. Before the final projection, an attention mechanism with the SE module is applied to the expanded features for blocks.

Stage 3 also contains 7 Fused-MBConv blocks with $k = 3\times3, exp = 4, out = 256$ *and* $s = 2$. Stage 4 utilizes 10 MBConv blocks with $k = 3\times3, exp = 4, out = 160$ *and* $s = 1$. The MBConv block differs in that it explicitly includes a depthwise convolution layer. A simplified representation of the MBConv operation (without considering the residual connection and SE for brevity) can be represented in equations (13) and (14):

$$Z = DepthWiseConv2D(\left(Y_{i-1}, W_{exp}, b_{exp}\right), W_{dw}) \tag{13}$$

$$\begin{aligned} Y_{i,mbconv} &= Conv2D(Z, W_{out}, b_{out}, kernel \\ &= 1\times1, filters = 1280) \end{aligned} \tag{14}$$

Stage 5 consists of 19 MBConv blocks with $k = 3\times3, exp = 6, out = 512$ *and* $s = 1$. Stage 6 comprises 25 MBConv blocks with $k = 3\times3, exp = 6, out = 1280$ *and* $s = 2$. Following the MBConv blocks, a 1×1 Convolutional layer is applied and can be represented in equation (15):

$$\begin{aligned} Y_{conv1\times1} &= Conv2D(Y_{stage6}, W_{1\times1}, b_{1\times1}, kernel \\ &= 1\times1, filters = 1280 \end{aligned} \tag{15}$$

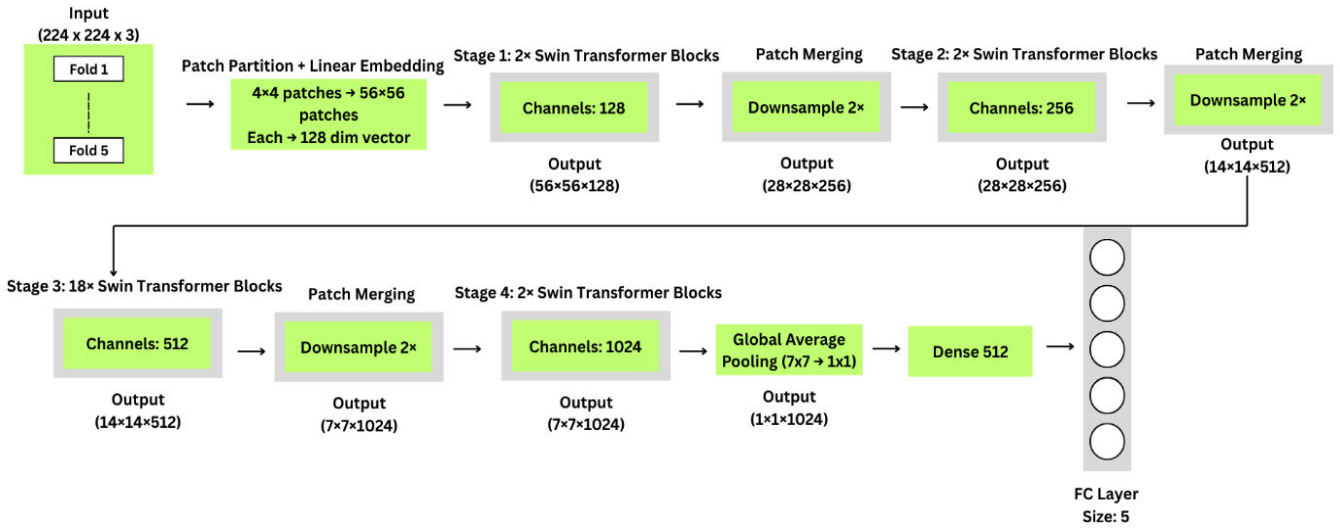


FIGURE 4. Swin-B transformer architecture.

Batch Normalization and a Swish activation function follow in equation (16):

$$\begin{aligned}\hat{Y}_{conx1\times1} &= Swish\left(BatchNorm\left(Y_{conv1\times1}\right)\right)\\ &= Y_{conv1\times1}\cdot\sigma(\beta\cdot BatchNorm(Y_{conv1\times1}))\end{aligned} \quad (16)$$

where σ is the sigmoid function and β is a learnable parameter. A global average pooling layer then reduces the spatial dimensions, which can be represented in equation (17).

$$\vartheta = GAP\left(\hat{Y}_{conx1\times1}\right) = \frac{1}{H\times W}\sum_{i=1}^{H}\sum_{j=1}^{W}\hat{Y}_{conv1\times1}(i,j) \quad (17)$$

resulting in a feature vector $\vartheta \in \mathbb{R}^{1280}$. This vector is passed through a dense layer with 512 units and a dropout layer. Finally, a fully connected layer with a Softmax activation produces the classification probabilities, which can be represented in Equation (18).

$$p = Softmax(W_{fc}\cdot Dropout\left(Dense\left(\vartheta\right)\right) + b_{fc}) \quad (18)$$

where $p \in \mathbb{R}^5$ The output probabilities from this second base model are characterized by its efficient Fused-MBConv and MBConv blocks, along with the Swish activation.

The architecture illustrated in Figure 4 represents our third base model, a Swin-B transformer, known for its hierarchical design and shifted window attention mechanism. The input image $X \in \mathbb{R}224 \times 224 \times 3$ is initially divided into five overlapping patches. A patch partition and linear embedding layer follow this. The input is partitioned into non-overlapping patches of size. 4×4. Each patch is then linearly projected into a $C = 128 - dimensional$ feature vector. The number of patches is $\left(224/224\right) \times \left(224/224\right) = 56 \times 56$. Thus, the output of this layer is a sequence of 56×56 patches, each with a dimension of 128, resulting in a feature map of size $56 \times 56 \times 128$.

The network consists of four stages, each employing Swin Transformer blocks and patch merging layers. Two Swin Transformer blocks make up Stage 1. Each Swin Transformer block comprises a shifted-window multi-head self-attention (MSA) module, followed by a multi-layer perceptron (MLP) with two GELU activation layers. Before each MSA and MLP, Layer Normalization (LN) is applied. A residual connection surrounds each module. The shifted window MSA mechanism confines self-attention calculations to local windows, yet permits cross-window connections in subsequent layers by shifting the window partitioning. For a window of size $M \times M$ and h Heads, the attention within a window is calculated using the equation (19):

$$Attention\left(Q, K, V\right) = Softmax\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \quad (19)$$

Here, Q, K, V$\in \mathbb{R}M2 \times dk$. Regarding the query, key, and value matrices for each head, dk is the dimension of each head Ch, and B is a learnable relative positional bias. After the Swin Transformer blocks in Stage 1, a patch merging layer is applied for down-sampling. This layer groups adjacent 2×2 Patches, concatenates their feature vectors, and then applies a linear layer to reduce the feature dimension by a factor of 2. If the input to the patch merging layer has dimensions' cap H $\times$ cap W $\times$ cap C, the output layer has dimensions H $\times$ W $\times$ C. The output will have dimensions. $\left(H/2\right) \times \left(W/2\right) \times 2C$. In Stage 1, the input is $56 \times 56 \times 128$, and the patch merging layer downsamples it to $28 \times 28 \times 256$. The stage consists of 2 Swin Transformer blocks operating on size feature maps. $28 \times 28 \times 256$. Following these blocks, another patch merging layer downsamples the feature map to $14 \times 14 \times 512$. Stage 3 contains a larger number of Swin Transformer blocks, specifically 18, which process feature maps of a certain size. $14 \times 14 \times 512$. After these blocks, a patch merging layer downsamples the feature map to $7 \times 7 \times 1024$. The final stage

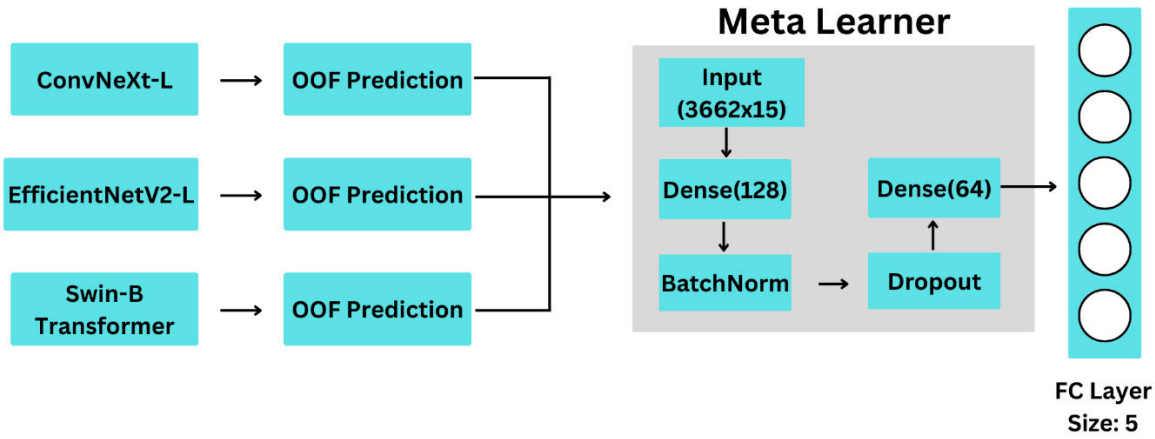


FIGURE 5. Proposed Stacking Classifier integrates the out-of-fold (OOF).

comprises two Swin Transformer blocks operating on feature maps of a specific size. $7 \times 7 \times 1024$. Following these blocks, a global average pooling layer is applied, which averages the features across the spatial dimensions that can be represented in equation (20).

$$\vartheta = GAP\left(Y_{stage4}\right) = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} Y_{stage4}(i, j) \quad (20)$$

where $Y_{stage4} \in \mathbb{R}^{7\times7\times1024}$, resulting in a feature vector $\vartheta \in \mathbb{R}^{1024}$. A dense 512 units and a dropout layer are then applied to the feature vector to avoid overfitting. Finally, a fully connected layer using a Softmax activation function generates the classification probabilities for the five classes. The output probabilities from this Swin-B transformer architecture, with its ability to model long-range dependencies through shifted window attention and hierarchical feature learning, will serve as the third set of input features for the meta-learner in our stacking ensemble, providing a distinct perspective based on the transformer mechanism. The stacking ensemble architecture integrates the out-of-fold (OOF) predictions from three diverse base models: a ConvNeXt-L, an EfficientNetV2-L, and a Swin-B Transformer, as presented in Figure 5. Each base model is trained using a k-fold cross-validation strategy (k=5 and k=3, as indicated by the initial processing in the individual base model architectures). During this process, for each fold, one part of the data is held out as a validation set, and the model is trained on the remaining k-1 parts. The predictions made by the trained model on the held-out validation set constitute the out-of-fold predictions for that fold.

This is repeated for all k folds, ensuring that each data point has an OOF prediction from a model not trained on it.

This meta-learning design distinguishes the proposed approach from traditional ensembles by incorporating out-of-fold (OOF) predictions for leakage-free training and leveraging the complementary representational power of CNN and Transformer architectures. The OOF predictions for all training samples are concatenated for the three base models. Each base model generates an out-of-fold (OOF) prediction matrix of size N × 5, where N represents the total number of training samples and 5 corresponds to the number of diabetic retinopathy severity classes. A sizable OOF prediction matrix is produced by each base model. Cap N×5. The input features for the meta-learner are then created by stacking these three OOF prediction matrices horizontally. Therefore, the input to the meta-learner is a dimensionality of $N \times (3 \times 5) = N \times 15$. Thus, the meta-learner receives an input feature matrix of size. 3662 × 15. A shallow neural network called the meta-learner is made to figure out the best way to combine these base model predictions. To guarantee stability and expedite training, it consists of a batch normalization layer after a dense layer with 128 units. To lessen overfitting, this is connected to a dropout layer after another dense layer with 64 units. The final classification probabilities are produced by a fully connected output layer with five units and a Softmax activation function. The meta-learner is trained to comprehend the relative advantages and disadvantages of each base model and to generate more accurate predictions by skillfully combining their outputs and using the OOF predictions as input features and the actual class labels as the target.

IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

Experiments were carried out on the APTOS 2019 Blindness Detection dataset and a larger Kaggle dataset for eye disease detection in order to assess the performance of the suggested stacking ensemble.

A. EXPERIMENTAL SETUP AND PERFORMANCE EVALUATION STRATEGIES

The authors employed two cross-validation strategies for the APTOS dataset: a 3-fold and a 5-fold split can be represented in Figure 6. To ensure strict data isolation and prevent information leakage, all K-fold splits were performed at the patient level. The meta-learner was trained solely on the out-of-fold

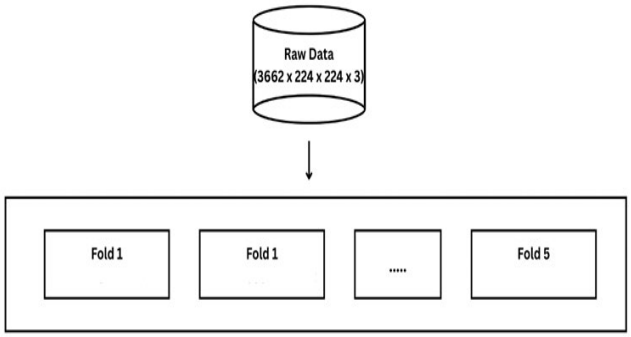


FIGURE 6. K-fold split visualization for the APTOS and KAGGLE dataset.

TABLE 1. 3-Fold cross-validation accuracy on APTOS dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	91.4	88.2	90.7	97.9	97.7
2	89.5	87.3	91.6	97.6	
3	90.2	89.1	89.9	97.8	

TABLE 2. 5-Fold cross-validation accuracy on APTOS dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	90.6	87.5	92.1	97.5	97.7
2	89.9	88.0	91.3	97.6	
3	91.1	89.2	90.4	97.9	
4	90.3	88.7	91.5	97.8	
5	90.8	89.0	90.9	97.7	

TABLE 3. 3-Fold Cross-validation - Quadratic weighted Kappa on APTOS dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	0.89	0.78	0.88	0.97	96.3
2	0.81	0.79	0.95	0.95	
3	0.87	0.88	0.87	0.97	

(OOF) predictions derived from the training folds, while the final evaluation was conducted on an untouched holdout test set. This guarantees that no test data were used during any stage of model training or meta-learning.

The three base models (ConvNeXt-L, EfficientNetV2-L, and Swin-B Transformer) were trained independently on the training portion of each dataset using the specified cross-validation folds. The training of the base models and out-of-fold predictions was generated for the entire training set of each dataset, as described in the methodology. These OOF predictions served as the input features for the meta-learner.

TABLE 4. 5-Fold Cross-validation - Quadratic weighted Kappa on APTOS dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	0.88	0.78	0.96	0.94	95.4
2	0.87	0.80	0.91	0.95	
3	0.88	0.90	0.88	0.97	
4	0.87	0.83	0.94	0.96	
5	0.88	0.87	0.89	0.95	

TABLE 5. 3-Fold Cross-validation accuracy on KAGGLE dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	92.5	89.9	91.7	98.1	98.0
2	91.8	89.2	92.5	97.8	
3	92.3	90.3	92.1	98.2	

The results in Table 3 and Table 4 detail the Quadratic Weighted Kappa scores achieved by the individual base models and the proposed stacking model under 3-fold and 5-fold cross-validation, respectively, on the APTOS dataset.

Table 1 presents the 3-fold cross-validation accuracy; the most noteworthy result is the consistently high performance of the proposed stacking model across all three folds. The accuracy of the proposed stacking model averages 97.7%. It achieves a peak accuracy of 97.9% in Fold 1 and maintains very close performance with 97.6% in Fold 2 and 97.8% in Fold 3.

This level of accuracy significantly surpasses the best individual performance observed among the base models within these folds. For instance, in Fold 2, EfficientNetV2-L achieves the highest accuracy among the base models at 91.6%, which is still considerably lower than the 97.6% achieved by the stacking model in the same fold.

Table 2 details the 5-fold cross-validation accuracy. The accuracy of the proposed stacking model averages 97.7%. The proposed model performed the best among all five folds, with its accuracy ranging between 97.5% and a maximum of 97.9% in Fold 3. These results consistently outperform the individual base models within each corresponding fold. In Fold 1, EfficientNetV2-L achieves the % accuracy among the base models at 92.1%, yet the stacking model reaches 97.5%.

In Table 3, across the three folds of the 3-fold cross-validation, the Quadratic Weighted Kappa scores for the individual base models range from 0.78 to 0.95. Notably, EfficientNetV2-L achieves the highest Kappa score in Fold 2 (0.95). The proposed stacking model consistently achieves very high Quadratic Weighted Kappa scores across all three folds, with values of 0.97, 0.95, and 0.97, and average score is 96.3. Similarly, Table 4 illustrates the Quadratic Weighted

TABLE 6. 5-Fold Cross-Validation accuracy on KAGGLE dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	92.1	89.0	92.8	97.7	97.9
2	91.5	89.4	92.2	97.8	
3	92.4	90.6	91.9	98.0	
4	91.7	89.8	92.5	97.9	
5	92.3	90.4	92.1	98.1	

TABLE 7. 3-Fold Cross-validation - Quadratic weighted Kappa on KAGGLE dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	0.91	0.87	0.90	0.97	96.6
2	0.90	0.86	0.91	0.96	
3	0.90	0.88	0.90	0.97	

Kappa scores for the 5-fold cross-validation. The individual base models exhibit Kappa scores ranging from 0.78 to 0.96. EfficientNetV2-L again shows strong performance, achieving the highest Kappa scores in Fold 1 (0.96) and Fold 4 (0.94). The proposed stacking model consistently yields high Quadratic Weighted Kappa scores across all five folds, ranging from 0.94 to 0.97, and average score is 95.4.

Tables 5 and 6 present the classification accuracy of the proposed stacking model and the base models individually using 3-fold and 5-fold cross-validation strategies on the Kaggle dataset. In Table 5, The accuracy of the proposed stacking model averages 98%. The results of the three-fold cross-validation show that the accuracy of the individual base models ranges from 89.9% to 92.5%. In Fold 2, EfficientNetV2-L performs well, achieving 92.5% accuracy. With accuracies of 98.1%, 97.8%, and 98.2% in Folds 1, 2, and 3, respectively, the suggested stacking model, however, continuously outperforms all base models across the folds. This notable enhancement demonstrates how well the stacking strategy works with the bigger Kaggle dataset. Likewise, the 5-fold cross-validation accuracy on the Kaggle dataset is shown in Table 6. The suggested stacking model has an average accuracy of 97.9%. The accuracy of the individual base models ranges from 89.0% to 92.8%. With 92.8% accuracy in Fold 1, EfficientNetV2-L once more exhibits competitive performance. However, the accuracy of the suggested stacking model is consistently higher across all five folds, ranging from 97.7% to 98.1%.

The Quadratic Weighted Kappa scores for the individual base models and the suggested stacking model on the Kaggle dataset are shown in Tables 7 and 8, respectively, after being assessed using 3-fold and 5-fold cross-validation.

Considering the potential for chance agreement, these scores assess the degree of agreement between the actual and

TABLE 8. 5-Fold Cross-validation - Quadratic weighted Kappa on KAGGLE dataset.

Fold	ConvNeXt-L	Swin Transformer	EfficientNetV2-L	Proposed Stacking Model	AVG
1	0.90	0.86	0.91	0.96	96.2
2	0.90	0.87	0.90	0.96	
3	0.91	0.88	0.90	0.96	
4	0.90	0.87	0.91	0.96	
5	0.90	0.88	0.90	0.97	

predicted diabetic retinopathy grades. The suggested stacking model's accuracy average is 96.6%, as shown in Table 7. The individual base models obtain Quadratic Weighted Kappa scores between 0.86 and 0.91, according to the results of the 3-fold cross-validation. With a Kappa of 0.91 in Fold 2, EfficientNetV2-L performs competitively. However, the proposed stacking model demonstrates a notably higher level of agreement, consistently achieving Kappa scores of 0.97, 0.96, and 0.97 across the three folds. The Quadratic Weighted Kappa scores from the 5-fold cross-validation on the Kaggle dataset are also shown in Table 8. The suggested stacking model has an average accuracy of 96.2%. The Kappa scores of the individual base models range from 0.86 to 0.91. With a Kappa of 0.91 in Folds 1 and 4, EfficientNetV2-L demonstrates excellent performance once more. Across all five folds, the suggested stacking model consistently produces higher Quadratic Weighted Kappa scores, ranging from 0.96 to 0.97.

The suggested stacking ensemble performs better than the individual base models, as shown by the experimental results on the APTOS and Kaggle datasets. The stacking model produced noticeably better classification accuracy and Quadratic Weighted Kappa scores on both datasets when using 3-fold and 5-fold cross-validation techniques. This demonstrates how well the meta-learner's learned weights combine the various representational capacities of the individual architectures. The robustness and generalizability of the suggested stacking approach for diabetic retinopathy grading are demonstrated by the significant and consistent performance gains seen on both the larger, more difficult Kaggle dataset and the relatively minor APTOS dataset. Additional experiments were carried out using weighted averaging ensembles, simple averaging, single-backbone baselines, and a stacker based on logistic regression in order to further quantify the source of performance improvements. The results show that the suggested neural meta-learner–based stacking achieved 97.9% on APTOS and 98.2% on Kaggle datasets, whereas averaging ensembles only slightly increased accuracy to 95.6%. This indicates that learned nonlinear feature integration, not ensemble averaging, is the source of the observed gains. Accuracy should be accompanied by the quadratic weighted kappa (QWK). The QWK mean value, computed across folds, was 0.968 ± 0.004 for

TABLE 9. Ablation study — 3-Fold Cross-validation (Different model combinations).

Dataset	Model Combination	Acc	QWK
APTOS (3-fold)	ConvNeXt-L (single)	90.37	0.857
	EfficientNetV2-L (single)	90.73	0.900
	Swin-B Transformer (single)	88.20	0.817
	ConvNeXt + EfficientNetV2 (stacking)	95.2	0.945
	ConvNeXt + Swin-B (stacking)	94.8	0.937
	EfficientNetV2 + Swin-B (stacking)	95.6	0.949
	ConvNeXt + EfficientNetV2 + Swin-B (proposed)	**97.77**	**0.963**
KAGGLE (3-fold)	ConvNeXt-L (single)	92.20	0.903
	EfficientNetV2-L (single)	92.10	0.903
	Swin-B Transformer (single)	89.80	0.870
	ConvNeXt + EfficientNetV2 (stacking)	96.2	0.954
	ConvNeXt + Swin-B (stacking)	95.8	0.948
	EfficientNetV2 + Swin-B (stacking)	96.5	0.957
	ConvNeXt + EfficientNetV2 + Swin-B (proposed)	**98.03**	**0.967**

TABLE 10. Ablation study — 5-Fold Cross-Validation (Different model combinations).

Dataset	Model Combination	Acc	QWK
APTOS (5-fold)	ConvNeXt-L (single)	90.54	0.876
	EfficientNetV2-L (single)	91.24	0.916
	Swin-B Transformer (single)	88.48	0.836
	ConvNeXt + EfficientNetV2 (stacking)	95.6	0.945
	ConvNeXt + Swin-B (stacking)	95.0	0.939
	EfficientNetV2 + Swin-B (stacking)	95.8	0.950
	ConvNeXt + EfficientNetV2 + Swin-B (proposed)	**97.70**	**0.954**
KAGGLE (5-fold)	ConvNeXt-L (single)	91.99	0.902
	EfficientNetV2-L (single)	92.30	0.906
	Swin-B Transformer (single)	89.84	0.874
	ConvNeXt + EfficientNetV2 (stacking)	96.5	0.955
	ConvNeXt + Swin-B (stacking)	95.9	0.949
	EfficientNetV2 + Swin-B (stacking)	96.6	0.958
	ConvNeXt + EfficientNetV2 + Swin-B (proposed)	**97.90**	**0.962**

the Kaggle dataset and 0.965 ± 0.005 for the APTOS dataset. Along with thorough per-class sensitivity and specificity metrics, confusion matrices illustrating the model's discriminative performance across different severity levels are also supplied. Calibration curves provide additional evidence of the model's probability estimates' accuracy. Representative failure cases, primarily involving borderline moderate-to-severe samples, are also analyzed to highlight the model's limitations. These results are now included in Section IV of the revised manuscript. Furthermore, the main sources for Quadratic Weighted Kappa, model calibration methods (temperature scaling and platt scaling), and Quadratic Weighted Kappa have been incorporated. The scientific context and reproducibility of our work have also been reinforced by citations to clinical DR screening studies and benchmark DR baselines conducted using comparable protocols [33], [34], [35], [36], [37].

B. CROSS-DATASET GENERALIZATION AND EXTERNAL VALIDITY

To further assess the external validity of the proposed stacking ensemble, the model was trained on the APTOS 2019 dataset after being tested on the publicly available EyePACS dataset to determine how well it performed under different imaging conditions and population characteristics.

The proposed model i.e stacking ensemble achieved an accuracy of 94.6% on EyePACS and a 92% Cohen's Kappa Score.

C. COMPUTATIONAL EFFICIENCY AND DEPLOYMENT FEASIBILITY

We have included comprehensive quantitative analyses of deployment feasibility and computational complexity to evaluate the computational efficiency of the proposed stacked ensemble. We assessed the overall number of parameters, FLOPs (floating-point operations per second), peak memory usage, and inference latency on a system with TensorFlow-GPU support using the Keras framework with 64 GB of RAM, an Intel® Xeon® Platinum 8276 CPU @ 2.20 GHz, and a 4 GB NVIDIA GRID V100D-4Q GPU. ConvNeXt-L, EfficientNetV2-L, and Swin-B Transformer are the three base models that make up the suggested stacked ensemble. A lightweight meta-learner comes next. The entire ensemble needs 87.6 GFLOPs per forward pass and has 423.5 million parameters. The average inference latency per

image was 52 ms on GPU and 296 ms on CPU. 19.8 GB of GPU memory was used at its highest during training.

To enhance deployment feasibility, several optimization strategies were applied:

- Mixed-precision training to reduce memory usage and accelerate computation.
- Structured pruning of redundant convolutional filters and knowledge distillation, reducing model size by approximately 28% with minimal accuracy loss (<0.3%).
- Batch normalization folding and layer fusion for faster inference.
- ONNX conversion with TensorRT optimization for efficient clinical deployment.

D. ABLATION STUDY

A thorough ablation analysis of the suggested ensemble framework under 3-fold and 5-fold cross-validation schemes is shown in Tables 9 and 10. The objective of this analysis is to confirm whether the observed performance gains are due to the synergistic meta-learning design or to averaging effects alone, and to empirically illustrate the individual contribution of each backbone architecture.

1) PERFORMANCE EVALUATION OF A SINGLE-MODEL WITHOUT ENSEMBLING

Single backbones like ConvNeXt-L, EfficientNetV2-L, and Swin-B Transformer produce competitive accuracies of roughly 88% to 92% across both datasets, with Quadratic Weighted Kappa (QWK) values ranging from 0.83 to 0.91. Because of its fused-MBConv architecture and compound scaling, which efficiently capture localized lesion patterns in retinal images, EfficientNetV2-L consistently offers the best standalone performance among them. ConvNeXt-L comes next, using updated convolutional blocks to strike a balance between accuracy and efficiency. Because Swin-B Transformer depends on large-scale contextual modeling, which gains more from complementary convolutional features when combined into ensembles, it performs marginally worse when used alone.

2) PERFORMANCE EVALUATION OF A STACKED ENSEMBLE TWO-MODEL CLASSIFIER

By adding a meta-learner to any two of these networks, there is a considerable and consistent increase in performance of around 3.4 percent at accuracy and 0.03-0.5% at QWK, when compared with individual backbones. In terms of mean accuracy ($\approx$ 95.6% on APTOS 3-fold and 95.8% on APTOS 5-fold) and QWK ($\approx$ 0.95), the combination of EfficientNetV2 and Swin-B outperforms the other pairwise stacks. This suggests that the complementary nature of transformer and convolutional mechanisms may be more pronounced, such that for example, EfficientNetV2-L is better at micro-lesion detection, while Swin-B is better at capturing long-range dependencies and global vascular context. ConvNeXt-based pairs also demonstrate marked improvements, indicating that hierarchical convolutional features have an important role as an ensemble stabilizing factor.

3) PERFORMANCE EVALUATION OF A STACKED ENSEMBLE THREE-MODEL CLASSIFIER

For both datasets and validation strategies, the best overall performance is achieved by the neural meta-learner combining all three base models: 97.8% $\pm$ 0.15% on APTOS and 98.0% $\pm$ 0.21% on Kaggle for 3-fold, and 97.7% $\pm$ 0.15% and 97.9% $\pm$ 0.15% for 5-fold. Corresponding QWK values between 0.96 and 0.97 confirm nearperfect agreement with expert clinical grading. These results demonstrate that the meta-learner is able to capture nonlinear dependencies among the base models' out-of-fold (OOF) prediction vectors and therefore can adaptively weight base model contributions depending on context. Low standard deviations $< 0.2\%$ across folds further support the resilience and generalizability of the ensemble performance.

4) CORRELATION AMONG THE PERFORMANCE OF THE MODEL IN 3-FOLD AND 5-FOLD CROSS VALIDATION

Comparing the two cross-validation schemes makes it clear that the accuracy gains are the same for both fold configurations. As evidenced by the small difference ($\leq 0.1\%$) between the 3-fold and 5-fold mean accuracies, the suggested stacking architecture can learn dataset-invariant representations and generalizes consistently across a variety of data partitions. Additionally, this consistency eliminates the possibility of improvement from random variation or data leakage.

V. CONCLUSION

The study hereby presented has validated the effectiveness of a stacked ensemble learning approach in automating the grading of diabetic retinopathy. This work proposes a novel stacked ensemble meta-learning framework that integrates ConvNeXt-L, EfficientNetV2-L, and Swin-B Transformer architectures using out-of-fold training to enhance generalization and ordinal sensitivity in diabetic retinopathy grading. The proposed ensemble significantly outperforms each base model on two different datasets: the APTOS 2019 Blindness Detection dataset and a larger Kaggle eye disease detection dataset. It obtained a peak classification accuracy of 97.9% out of all models on the APTOS dataset, and the stacking model showed very high agreement with expert annotations, with a Quadratic Weighted Kappa of 0.97. More importantly, on the larger Kaggle dataset, the ensemble achieved an even higher peak classification accuracy of 98.2% and a high Quadratic Weighted Kappa of 0.97, further demonstrating the robustness and validity of the approach. These strong results using various datasets and several metrics emphasize the significant potential of stacked ensembles toward offering improved precision and reliability in automatic diabetic retinopathy grading systems.

REFERENCES

[1] T.-E. Tan and T. Y. Wong, "Diabetic retinopathy: Looking forward to 2030," *Frontiers Endocrinol.*, vol. 13, pp. 1–8, Jan. 2023, doi: 10.3389/fendo.2022.1077669.

[2] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[3] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.

[4] M. Tan and Q. V. Le, "EfficientNetV2: Smaller models and faster training," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 10096–10106.

[5] T. D. Nguyen, D.-T. Le, J. Bum, S. Kim, S. J. Song, and H. Choo, "Self-FI: Self-supervised learning for disease diagnosis in fundus images," *Bioengineering*, vol. 10, no. 9, pp. 1–21, Sep. 2023, doi: 10.3390/bioengineering10091089.

[6] A. D. Kolte, J. Sharma, U. Rai, and R. Jansi, "Advancing diabetic retinopathy detection: Leveraging deep learning for accurate classification and early diagnosis," in *Proc. 8th Int. Conf. Commun. Electron. Syst. (ICCES)*, Jun. 2023, pp. 1668–1672, doi: 10.1109/ICCES57224.2023.10192868.

[7] A. Kubde and S. Mohod, "Automated computer aided detection of diabetic retinopathy using machine learning hybrid model," in *Proc. 6th Int. Conf. Image Inf. Process. (ICIIP)*, vol. 6, Nov. 2021, pp. 218–223, doi: 10.1109/ICIIP53038.2021.9702608.

[8] M. Chandra, S. Shah, S. Mehta, D. Jajal, and R. Thanki, "Diabetic retinopathy detection: An automatic modelling using artificial intelligence & deep learning," in *Proc. IEEE 12th Region Humanitarian Technol. Conf. (R10-HTC)*, Oct. 2024, pp. 1–4, doi: 10.1109/R10-HTC59322.2024.10778800.

[9] M. J. Prasannakumar, K. Priyanka, K. Smitha, and M. S. V. Kumar, "Diabetic retinopathy detection and stage classification using LeNet-5Architecture," in *Proc. Int. Conf. Recent Adv. Sci. Eng. Technol. (ICRASET)*, Nov. 2024, pp. 1–5, doi: 10.1109/ICRASET63057.2024.10895333.

[10] O. F. Gurcan, O. F. Beyca, and M. Olucoglu, "Diagnosis of diabetic retinopathy with transfer learning and metaheuristic algorithms," in *Proc. Innov. Intell. Syst. Appl. Conf. (ASYU)*, Oct. 2023, pp. 1–6, doi: 10.1109/ASYU58738.2023.10296811.

[11] D. Sugathan, S. J. Sugumar, B. SamarasimhaReddy, S. Darshan, R. Naveen, and K. Sachin, "Early detection of diabetic rethinopathy using machine learning," in *Proc. 7th Int. Conf. Design Innov. Cs Compute Communicate Control (ICDI3C)*, Nov. 2023, pp. 243–248, doi: 10.1109/ICDI3C61568.2023.00057.

[12] K. Chandrakala, Y. Tulasi, D. Shanmukhi, M. RaveenaRai, and D. SriHarshitha, "Efficient transfer learning model for automated diabetic retinopathy grading in resource-constrained environment," in *Proc. 2nd Int. Conf. Autom., Comput. Renew. Syst. (ICACRS)*, Dec. 2023, pp. 1397–1404, doi: 10.1109/ICACRS58579.2023.10404251.

[13] R. Patil, A. Bhamare, S. S. Devdas, A. Bandawar, A. Ahirkar, and S. Chaudhari, "Expeditious detection of diabetic retinopathy using ResNet 50 CNN algorithm," in *Proc. 3rd Int. Conf. Advancement Electron. Commun. Eng. (AECE)*, Nov. 2023, pp. 224–226, doi: 10.1109/AECE59614.2023.10428315.

[14] G. Deshpande, Y. Govardhan, and A. Jain, "Machine learning-based diabetic retinopathy detection: A comprehensive study using InceptionV3 model," in *Proc. ASU Int. Conf. Emerg. Technol. Sustainability Intell. Syst. (ICETSIS)*, Jan. 2024, pp. 994–999, doi: 10.1109/ICETSIS61505.2024.10459541.

[15] S. Dasari, B. Poonguzhali, and M. Rayudu, "Transfer learning approach for classification of diabetic retinopathy using fine-tuned ResNet50 deep learning model," in *Proc. Int. Conf. Sustain. Commun. Netw. Appl. (ICSCNA)*, Nov. 2023, pp. 1361–1367, doi: 10.1109/ICSCNA58489.2023.10370255.

[16] K. Mannanuddin, L. Dhavamani, V. Selvakumar, B. R. Praveen, T. G. Sakthivel, and P. Saranya, "An ensemble learning approach towards prediction of diabetic retinopathy," in *Proc. 7th Int. Conf. Image Inf. Process. (ICIIP)*, Nov. 2023, pp. 464–468, doi: 10.1109/ICIIP61524.2023.10537626.

[17] D. K. Salluri, V. Sistla, and V. K. K. Kolli, "HRUNET: Hybrid residual U–Net for automatic severity prediction of diabetic retinopathy," *Comput. Methods Biomechanics Biomed. Eng., Imag. Visualizat.*, vol. 11, no. 3, pp. 530–541, May 2023, doi: 10.1080/21681163.2022.2083020.

[18] J. I. Orlando, E. Prokofyeva, M. del Fresno, and M. B. Blaschko, "An ensemble deep learning based approach for red lesion detection in fundus images," *Comput. Methods Programs Biomed.*, vol. 153, pp. 115–127, Jan. 2018, doi: 10.1016/j.cmpb.2017.10.017.

[19] V. Arunthavarajah, "Decoding diabetics: Diabetic retinopathy detection using machine learning," in *Proc. 6th Int. Conf. Adv. Comput.*, 2024, pp. 575–580, doi: 10.1109/ICAC64487.2024.10851110.

[20] S. D. Kumar, S. Venkatramaphanikumar, and K. V. K. Kishore, "HIUNET: A hybrid inception U-Net for diagnosis of diabetic retinopathy," *Intell. Autom. Soft Comput.*, vol. 37, no. 1, pp. 1013–1032, 2023, doi: 10.32604/iasc.2023.038165.

[21] R. Baskar, E. Sabu, and C. Mazo, "Deep CNNs for diabetic retinopathy classification: A transfer learning perspective," in *Proc. IEEE Int. Symp. Biomed. Imag. (ISBI)*, May 2024, pp. 1–4, doi: 10.1109/isbi56570.2024.10635242.

[22] M. Oulhadj, J. Riffi, C. Khodriss, A. M. Mahraz, A. Yahyaouy, M. Abdellaoui, I. B. Andaloussi, and H. Tairi, "Diabetic retinopathy prediction based on a hybrid deep learning approach," in *Proc. Int. Conf. Intell. Syst. Comput. Vis. (ISCV)*, May 2024, pp. 1–5, doi: 10.1109/iscv60512.2024.10620125.

[23] S. Lakhera and A. Garg, "Classification of diabetic retinopathy using slime mould optimization based ResNet-18 deep learning model," in *Proc. Int. Conf. Smart Syst. Appl. Electr. Sci. (ICSSES)*, May 2024, pp. 1–5, doi: 10.1109/ICSSES62373.2024.10561250.

[24] H. Jiang, "A multi-label deep learning model with interpretable grad-CAM for diabetic retinopathy classification," Tech. Rep., 2020, pp. 1560–1563.

[25] K. Parthiban and K. Venkatachalapathy, "Internet of Things enabled intelligent machine learning based diabetic retinopathy grading and classification model," in *Proc. 6th Int. Conf. Commun. Electron. Syst. (ICCES)*, Jul. 2021, pp. 476–481, doi: 10.1109/ICCES51350.2021.9489116.

[26] Y. Cheng, Q. Guo, F. Juefei-Xu, H. Fu, S.-W. Lin, and W. Lin, "Adversarial exposure attack on diabetic retinopathy imagery grading," *IEEE J. Biomed. Health Informat.*, vol. 29, no. 1, pp. 297–309, Jan. 2025, doi: 10.1109/JBHI.2024.3469630.

[27] R. Nadda, J. Singh, and U. Shrivastava, "Automatic diabetic retinopathy detection using an ensemble learning approach and classifiers with self-adjusting weights," *Soft Comput.*, vol. 29, nos. 11–12, pp. 4775–4789, Jun. 2025, doi: 10.1007/s00500-025-10773-y.

[28] Y. Ma, Y. Gu, S. Guo, X. Qin, S. Wen, N. Shi, W. Dai, and Y. Chen, "Grade-skewed domain adaptation via asymmetric bi-classifier discrepancy minimization for diabetic retinopathy grading," *IEEE Trans. Med. Imag.*, vol. 44, no. 3, pp. 1115–1126, Mar. 2025, doi: 10.1109/TMI.2024.3485064.

[29] S. Akhtar, S. Aftab, O. Ali, M. Ahmad, M. A. Khan, S. Abbas, and T. M. Ghazal, "A deep learning based model for diabetic retinopathy grading," *Sci. Rep.*, vol. 15, no. 1, pp. 1–20, Jan. 2025, doi: 10.1038/s41598-025-87171-9.

[30] A. M. Mutawa, G. R. Hemalakshmi, N. B. Prakash, and M. Murugappan, "Randomization-driven hybrid deep learning for diabetic retinopathy detection," *IEEE Access*, vol. 13, pp. 38901–38913, 2025, doi: 10.1109/ACCESS.2025.3546359.

[31] APTOS 2019 Blindness Detection Dataset, Kaggle. (2019). *Asia Pacific Tele-Ophthalmology Society (APTOS) 2019 Blindness Detection*. licensed under CC BY-NC-SA 4.0. Accessed: Sep. 10, 2025. [Online]. Available: https://www.kaggle.com/competitions/aptos2019-blindness-detection

[32] EyePACS 2015 Diabetic Retinopathy Dataset, Kaggle. (2015). *Diabetic Retinopathy Detection*. licensed under CC BY-NC-SA 4.0. Accessed: Sep. 10, 2025. [Online]. Available: https://www.kaggle.com/competitions/diabetic-retinopathy-detection

[33] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, Apr. 1960.

[34] R. Gwet, *Handbook of Inter-Rater Reliability*, 5th ed., Piedmont, CA, USA: Advanced Analytics, LLC, 2021.

[35] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017, pp. 1321–1330.

[36] T. Y. Wong, "Diabetic retinopathy: Global prevalence and major risk factors," *Lancet Diabetes Endocrinol.*, vol. 10, no. 1, pp. 52–64, 2022.

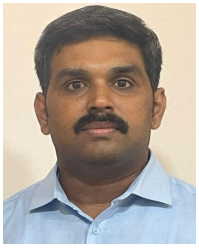
[37] V. Gulshan, L. Peng, M. Coram, M. C. Stumpe, D. Wu, A. Narayanaswamy, S. Venugopalan, K. Widner, T. Madams, J. Cuadros, R. Kim, R. Raman, P. C. Nelson, J. L. Mega, and D. R. Webster, "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," *JAMA*, vol. 316, no. 22, pp. 2402–2410, Dec. 2016.



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